**8** **(a)** State the formula for the de Broglie wavelength $\lambda$ of a moving particle.

State the meaning of any other symbol used.

.....

.....

..... [2]

**(b)** Electrons accelerate through a potential difference, pass through a thin crystal and are then incident on a fluorescent screen.

The pattern in Fig. 8.1 is observed on the fluorescent screen.

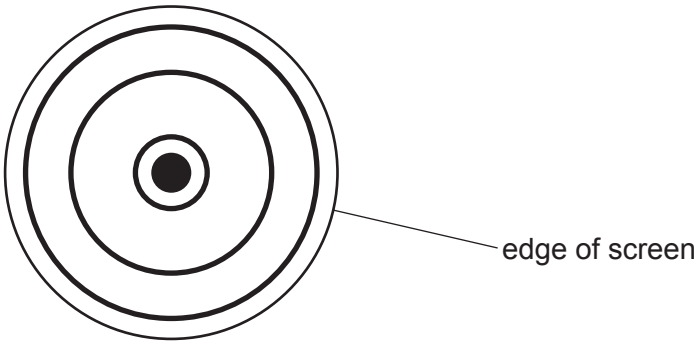



**Fig. 8.1** not to scale

**(i)** State the name of the phenomenon shown by the electrons at the crystal.

..... [1]

**(ii)** State what this phenomenon shows about the nature of electrons.

.....

..... [1]

**(iii)** Suggest why the thin crystal causes the phenomenon in **(b)(i)**.

.....

..... [1]

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**(iv)** The electron is accelerated through a different potential difference. The new pattern observed on the screen is shown in Fig. 8.2.

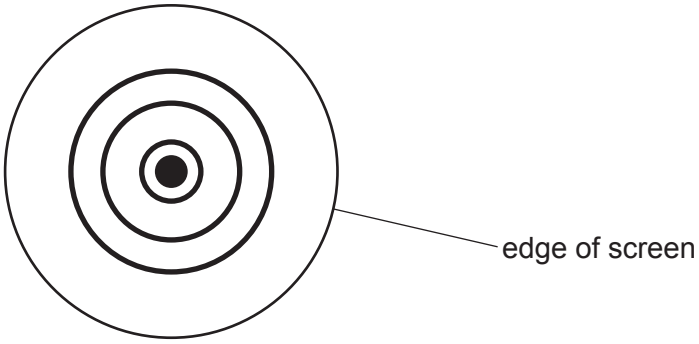



**Fig. 8.2** not to scale

State and explain the change that has been made to the potential difference to create the pattern shown in Fig. 8.2.

.....

.....

.....

..... [2]

[Total: 7]